DATA STRUCTURES AND ALGORITHMS

Lecture Notes 5.2

Pradit Pitaksathienkul

ROAD MAP

- TREES
  - Implementation of Trees
  - Tree Traversals
  - Binary Trees
  - **The Search Tree ADT-Binary SearchTrees**

Binary Search Trees

Each node in tree stores an item

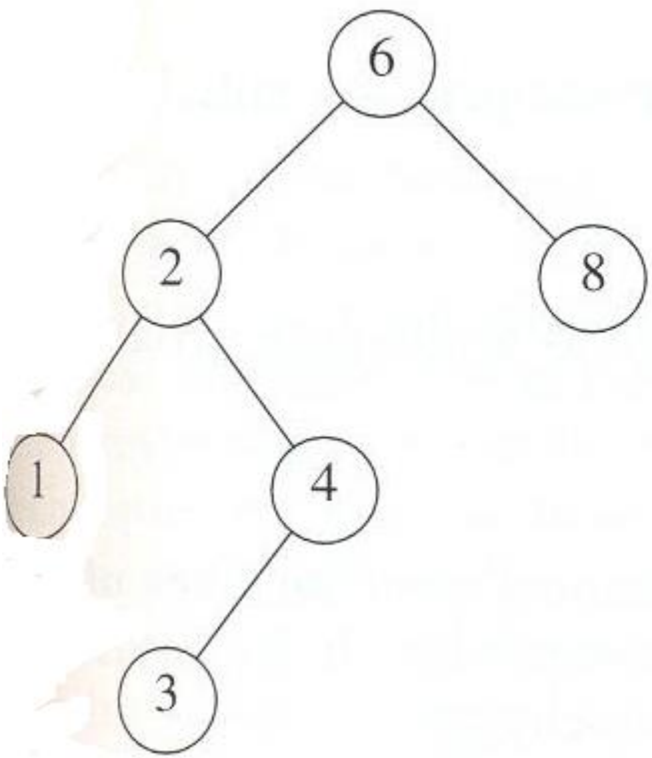
- items are integers and distinct
- when has no left or ritght child node, mean left or right point to None.

Properties

- A binary tree
- for each node ***x***
  - values at left subtree are smaller
  - values at right subtree are larger

  than the value of ***x***

find



Find min

```
/* Find min value node in the tree - Return the current node
   that minimum */

def _min_value_node(self, node):
    current = node
    while current.left is not None:
        current = current.left
    return current
```

- What else to find the max value?